

Provide & Inject

Dependency Injection in Vue

The Prop Drilling Problem

- Passing props through many component layers
- Intermediate components don't use the data
- Tedious and error-prone
- Makes refactoring difficult

Prop Drilling in Action

```
<!-- App.vue -->
<script setup lang="ts">
import { ref } from 'vue'
const theme = ref('dark')
</script>
<template>
  <Layout :theme="theme" />
</template>

<!-- Layout.vue - just passing through -->
<script setup lang="ts">
defineProps<{ theme: string }>()
</script>
<template>
  <Sidebar :theme="theme" />
  <Main :theme="theme" />
</template>
```

```
<!-- Sidebar.vue - still just passing -->
<script setup lang="ts">
defineProps<{ theme: string }>()
</script>
<template>
  <Navigation :theme="theme" />
</template>

<!-- Navigation.vue - finally uses it! -->
<script setup lang="ts">
const { theme } = defineProps<{ theme: string }>()
</script>
<template>
  <nav :class="theme">{{ theme }}</nav>
</template>
```

provide/inject to the Rescue!

- Parent provides a value
- Any descendant can inject it
- No intermediate props needed
- Works at any depth

```
<!-- App.vue -->  
<script setup lang="ts">  
import { provide, ref } from 'vue'  
  
const theme = ref('dark')  
provide('theme', theme)  
</script>  
  
<template>  
  <Layout /> <!-- No :theme prop needed! -->  
</template>
```

Injecting the Value

```
<!-- Navigation.vue - finally uses it! -->
```

```
<script setup lang="ts">
```

```
import { inject, type Ref } from 'vue'
```

```
const theme = inject<Ref<string>>>('theme')
```

```
</script>
```

```
<template>
```

```
  <nav :class="theme">{{ theme }}</nav>
```

```
</template>
```

Type-Safe provide/inject

```
// injection-keys.ts
import type { InjectionKey, Ref } from 'vue'

export const ThemeKey: InjectionKey<Ref<string>> = Symbol('theme')

// App.vue:      provide(ThemeKey, ref('dark'))
// Navigation.vue: const theme = inject(ThemeKey)
//               ^? Ref<string> | undefined
```

Reactive provide

```
<!-- App.vue -->
<script setup lang="ts">
import { provide, ref } from 'vue'
import { ThemeKey } from './injection-keys'

const theme = ref('dark')
provide(ThemeKey, theme)

function toggleTheme(): void {
  theme.value = theme.value === 'dark' ? 'light' : 'dark'
}
</script>

<template>
  <button @click="toggleTheme">Theme: {{ theme }}</button>
  <Layout /> <!-- Navigation will see updates! -->
</template>
```


Questions?

Have you used provide/inject to avoid prop drilling?

Thank You!

Remember: provide/inject beats prop drilling!